New Approach to Studying the Concept of Information for IT-Students

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Abstract. The concept of information is fundamental in the IT education. Understanding the meaning and significance of information is required by modern educational standards of higher school. However, in existing educational courses, as a rule, the problem of defining this term is ignored. Learning is based on intuitive ideas. There is a methodological contradiction: the development and application of information technologies requires accuracy and rigor, but all constructions are based on a vague, intuitive concept. The purpose of this work is to develop a new approach to studying the concept of information within the framework of IT-education. Here information is understood as the result of a subset selection from a set of elements. The significance of the proposed approach is revealed using the example of numerical methods of multimodal optimization. It is shown that the new approach makes it possible to formulate effective stopping conditions for a stochastic optimal search algorithm. This stopping criterion guarantees a given amount of information about the location of the global optimum. The developed approach was introduced into the educational process of training IT students within the academic disciplines "Information Theory", "Bio-inspired optimization methods" and "Mathematical modeling of selection processes".

Keywords: Information · Selection · Optimization · IT-education

1 Introduction

The needs of the modern information society dictate new tasks for the education system [1,2]. The most important of them is the formation of digital competencies [3,4]. The demands of today are no longer satisfied by the acquisition of "computer literacy", i.e. mastering basic computer skills. Now education should provide a higher level of qualifications – mastery of "digital culture". It presupposes conscious knowledge of information technologies, understanding of their theoretical foundations, a critical attitude towards their use, and the ability to work in a digital environment and with digital products. This task is especially relevant for training specialists in the area of information technology [5,6].

In the system of digital competencies, the basic, system-forming concept is the concept of information. It occupies a central place in a number of competencies defined by current educational standards [4,6].

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At the same time, in computer science textbooks, the fundamental concept of information is defined incorrectly. The concepts "information", "data", "knowledge", "messages", "signal" are often confused. In this case, the essence of the information is not revealed at all, only one vague concept is replaced by another, equally vague [7]. We can give several examples of such an explanation of the term "information" in various textbooks: "Information is the world around us, the exchange of signals between living and inanimate nature, people and devices" [8]; "information is a product of the interaction of data and methods adequate to them" [9]. The authors of such definitions do not realize that data is only a form of information representation for processing, storage and transmission, and knowledge is conscious information. Information is not defined using such concepts; on the contrary, these concepts themselves are defined through information. Sometimes it is argued that the notion of information is indefinable, just as the basic notions of "number" and "set" are not defined in mathematics. But this analogy is inappropriate, since indefinable notions of mathematics are introduced through a system of axioms that uniquely determine their properties in relation to each other. And no axioms are given for information.

Due to the need for a detailed understanding of the properties of information, the discipline "Information Theory" is usually included in the curricula of IT-students. Here the mathematical properties of information are studied on the base of probability theory. However, a strict definition of information is not given. The classical theory considers only its quantitative aspect. But it is obvious that we must distinguish between information and quantity of information. For example, the amount of information in the famous message "Edwardum occidere nolite (,) timere (,) bonum est" (Marlow's Edward II, Act 5, Scene IV) does not depend on where the comma is placed, but the information will change to the opposite. In addition, this approach depends significantly on the theory of probability and connects information with randomness. But this reduces the possibility of applying the concept of information in modern information technologies [10]. For example, this approach is difficult to use in geoinformatics, where, as a rule, information is not limited to random outcomes.

The noted problems and contradictions are explained by the fact that there is currently no scientific consensus regarding a strict definition of information [11]. There are many particular approaches to understanding it, which sometimes contradict each other [12–14]. New approaches are being developed, new concepts of information and its quantitative measurement are emerging [15]. Due to the lack of a universal definition, attempts are being made to provide a specific formalization of information in partial subject areas [16–18].

At the same time, there is an obvious methodological contradiction in teaching. The study of digital technologies requires formal rigor, logical consistency and validity, but the learning is based on vague, intuitive concepts. It is necessary to give a working definition of information to solve the problems of IT education. This definition may not be universal, but has practical significance in the area of information technology.

The purpose of this study is to develop a correct approach to understanding the term "information" for training IT-students. In addition, the objective of the work is to demonstrate the applied value of the proposed approach using global optimization methods as an example [19]. The optimization method is considered as a process of gradually obtaining more and more information about the location of the global optimum point. The new approach allows us to formulate a stopping condition for the algorithm, which in some cases guarantees obtaining the required amount of information with a given probability.

2 Materials and Methods

A new approach to understanding information is based on the mathematical procedure of selecting a certain subset from the original set of elements. To formalize the selection procedure, the apparatus of fuzzy sets is used [20].

The applied significance of the developed approach is analyzed using the example of bioinspired global optimization methods [21–23]. There is a problem of the correct condition for stopping the algorithm. The simplest condition is a limit on the number of steps. However, there is no guarantee that the solution obtained in a fixed number of steps will be a good approximation of the desired optimum. Another condition is stopping improvement of objective function values. If, starting from a certain point, the resulting solution remains practically unchanged, then the optimization process ends. But for heuristic methods there is a danger of stopping improvements due to hitting a local extremum [24]. A much more effective stopping condition is achieving a given accuracy of the optimum approximation that is used in some convergent deterministic search algorithms. However, in stochastic algorithms, it is usually impossible to evaluate the resulting approximation accuracy. The problem of constructing an effective stopping condition for the method remains relevant.

A relatively new and fairly simple stochastic method of multi-modal optimization SoFA (Survival of the Fittest Algorithm) is used to illustrate the effectiveness of the new approach in obtaining the stopping condition [25]. Its performance has been studied in detail on a number of relevant test functions, including special classes of randomly generated GKLS functions [26].

The method SoFA solves the following problem: to find the point of greatest value of a continuous non-negative function $F(z)$ defined in the hyperparallelepiped $\Pi = \{z = (x_1, ...x_n) : a_i \leq x_i \leq b_i, i = 1, ...n\}$; D is the length of the largest edge of Π. For simplicity, it is assumed that the function F has a single global maximum point in Π. First, a point $z_1 = (x_{11}, \dots x_{n1})$ is randomly selected from Π and the value of the function $F(z_1)$ is calculated. At each next step, another point is added in a certain way, at which the value of the function F is also calculated. To generate such a point, the following operations are performed at each k-th step of the algorithm:

1. For all points of previous tests, the values

$$\phi(i,k) = \frac{F^{\psi(k)}(z_i)}{F^{\psi(k)}(z_1) + ... + F^{\psi(k)}(z_{k-1})}$$

are calculated and are compared with a predetermined small positive parameter γ. Here $\psi(k)$ is a tunable variable monotonously depending on the step k, satisfying the condition $\psi(k) \to +\infty$ as $k \to \infty$; γ is the algorithm parameter. Based on the comparison results, the characteristic indicator $\mu(i,k)$ is determined as follows:

$$\mu(i,k) = 1, \quad if \quad \phi(i,k) \geq \gamma; \quad \mu(i,k) = 0, \quad if \quad \phi(i,k) < \gamma. \tag{1}$$

The set of points for which $\mu(i,k) = 1$ is called the population at the k-th step of the method (k-th generation of the population). Please note that the population cardinality changes from step to step. The idea of using a population of variable size in evolutionary algorithms was used in [27]. Note also that if at some step the indicator $\mu(i,k)$ for the i-th point becomes equal to zero, then it remains zero at all subsequent steps.

2. From the points z_j included in the population at the k-th step, one point is randomly selected as a reference point $w_k = (w_{1k}, \ldots, w_{nk})$ with the corresponding probabilities

$$p(j,k) = \frac{F^{\psi(k)}(z_j)}{\mu(1,k)F^{\psi(k)}(z_1) + \cdots + \mu(k-1,k)F^{\psi(k)}(z_{k-1})}.$$

3. Based on the reference point w_k, the next test point $z_k = (x_{1k}, \ldots x_{nk})$ is constructed. Each of its coordinates x_{ik} is randomly selected from the segment $[a_i, b_i]$ with a probability density $A_{ik}exp(-(x_{ik} - w_{ik})^2/\sigma_k)$. Here σ_k is the doubled variance of the normal distribution, depending on the step, the constant A_{ik} is chosen so that the normalization condition is satisfied, that is

$$A_{ik} = \frac{1}{\int_{a_i}^{b_i} \exp(-\frac{(x_{ik}-w_{ik})^2}{\sigma_k})dx_{ik}}. \tag{2}$$

4. At the selected point z_k, the value of the objective function $F(z_k)$ is calculated.

Then the process is repeated. The condition for stopping the method in the simplest case is the exhaustion of a given number of steps. The work constructs other more relevant conditions for stopping the method.

3 Results

3.1 Definition of Information

Within the framework of the new approach, information is understood as a selected subset in relation to the original set of admissible elements (alternatives, outcomes, states). In particular, only one alternative from the original set can be selected. This allows us to understand information in its essence, as opposed to its quantitative measurement.

Information is inextricably linked to reducing the number of alternatives. For example, suppose one card is randomly drawn from a standard deck containing 36 playing cards. If the suit of this card is known, then the set of outcomes is reduced from 36 to 9 cards of this suit. This corresponds to receiving 2 bits of information. But the information is not in this quantity. For different suits, the corresponding sets of outcomes will be different. This means that the information meaningfully corresponds not to a quantitative reduction of the set cardinality, but to the remaining subset itself. The idea of this approach was given by N. Wiener: "If we have a set and there is a subset of this set, then ... the information is contained in this subset" [28].

For a strict mathematical formalization of selection, it is necessary to introduce an indicator of the elements membership in the selected subset. In classical set theory, this indicator is determined by the characteristic function of a subset and can take only two values – zero or one. In the above SoFA algorithm, the indicator $\mu(i,k)$ is an indicator of whether the i-th test point belongs to the population at the k-th step, the function μ is the characteristic function of the population. For a broader understanding of selection, it is advisable to expand the classical construction and allow one to consider fuzzy subsets in the sense of L. Zadeh [20]. In this case, the membership indicator for the selected fuzzy subset can take intermediate values from zero to one. This value characterizes the degree to which a particular element belongs to the selected subset. Thus, the selected subset is actually specified by the distribution of the membership indicator over the original set of elements. For mathematical convenience, we can always go to such a distribution when the total sum of the membership indicators for all elements is equal to one. In particular, the probability of the outcome or the frequency of the state can be chosen as the indicator of membership. The selection of a subset from the original set of elements can be carried out not only on the base of probability theory. For example, in biological systems, the indicator of membership can be the proportion of the number of species in the community; in economic systems, it is specific demand for a product, etc. [3].

If we consider an ordinary distinct subset, we can numerically characterize the degree of its distinctness. In the case where the distribution is a normalized positive measure, the degree of salience is characterized by the measure of this distinct subset.

The concept of diversity is also of great importance in information theory. In the simplest case, diversity is the total number of equally likely outcomes. There are many other definitions and formulas for calculating diversity. Diversity characterizes the information capacity of the system. Obviously, when selecting one element from 100 equal alternatives, more information will be obtained than when selecting the same element from 10 alternatives.

Note that in the search optimization method, adding a new test point increases the diversity of the test set, if diversity is understood as the cardinality of the set. Consequently, the information capacity of this set increases, which potentially leads to more information about the position of the global optimum and a more accurate approximation of it.

The proposed explanation of information makes this concept quite strict and transparent for students perception. It allows us to solve a methodological problem in the formation of digital culture. Research results were introduced into the educational process at Lobachevsky State University of Nizhny Novgorod. The new approach to understanding information was used to modernize the academic discipline "Information Theory" in accordance with the updated requirements of educational standards [5]. This discipline is studied in the second year of a bachelors degree of the studying area "Fundamental Informatics and Information Technologies." By that time, students are already familiar with the basic mathematical facts and are ready to accept the formalization based on ideas about fuzzy sets, as well as the construction of the distribution of the membership index. Two hours of lectures are allocated from the total labor intensity of the discipline to cover the topic "What is information." Within its framework, the above strict definition and a number of motivational examples are considered. Students' learning of the material is monitored using developed electronic tests in the MOODLE environment.

It should be noted that the meaning of the new definition of information will not be fully revealed without demonstrating its significance for practical problems. Applied aspects of the new approach are discussed in the next training courses in specialized disciplines.

3.2 Diversity Assessment

Here we analyze the global optimization method to illustrate the role of a new understanding of information. Numerical global optimization methods can be understood as a process of gradually obtaining information about the location of the global optimum point in the search space. Finding this point accurately using numerical methods is virtually impossible. We can only specify the subset containing it. The information here is a distinguished subset containing the global extremum. As the number of steps of the method increases, this subset gradually narrows, which corresponds to more information being obtained. In addition, the population isolated at each step from a set of tests can be considered as information obtained by this time about points that are promising for further work.

Modern bioinspired global optimization methods combine two complementary processes - exploration and exploitation [30,31]. The exploration is aimed at studying the behavior of the objective function in the search space and selecting promising subareas. The exploitation consists of improving the result in promising local subareas. The first process can be understood as an increase in information capacity for the search system, the second - as making an improving decision based on this information capacity. It is necessary to effectively organize and coordinate these two processes. To do this, we need to have some characteristics that reflect their quality. We can evaluate the quality of the first process using the diversity of the entire test set or the diversity of the current population. However, if we understand diversity as the set cardinality, then this value will not characterize the quality of the search space exploration, since all

points can be concentrated in a narrow local subarea. A different understanding of diversity is needed.

Another estimation of population diversity in global optimization methods was proposed in [32]. In fact, here diversity refers to the dispersion of points included in the population. High diversity corresponds to a significant spread of population points. This allows us to more fully cover the search space with population points and prevents premature concentration of the population in the vicinity of the local optimum [33]. At the final steps of the method, when the exploitation process prevails, there is a significant decrease in population diversity. However, long-term preservation of a large diversity does not guarantee better results, since it can negatively affect the speed of convergence and accuracy of the solution found [34]. Thus, the introduced diversity assessment has only an indirect heuristic meaning. The question remains which values of diversity at which stages are necessary to effectively solve the problem.

A different understanding of diversity can be given. Let us divide the hyperparallelepiped Π into M identical cells that do not have common internal points. Let us denote the minimal edge length of each cell by 2θ. Then, by the diversity of the set of test points we mean the number of cells in which at least one test point fells. Accordingly, the diversity of a population is the number of cells in which at least one point of the population falls. The diversity corresponds to the number of those cells in which there is data on the behavior of the target function. Thus, diversity characterizes the proportion of the explored search space and the studied behavior of the function.

Let us show that the SoFA method guarantees the achievement of such diversity for any fixed value of the number M with a given probability for a sufficiently large number of steps.

Theorem 1. *Let*

$$\sigma_k = \frac{2nD^2}{\ln k}. \tag{3}$$

Then each cell will contain at least one test point generated using the SoFA in N steps with probability greater than

$$1 - M\big(1 - \big(\frac{\theta\sqrt{2\ln N}}{D\sqrt{\pi}}\big)^n\big(\frac{1}{\sqrt{N}}\big)\big)^N. \tag{4}$$

Proof. Lets take an arbitrary positive number θ and an arbitrary point $z = (x_1, \dots x_n) \in \Pi$. Consider its cubic θ-neighborhood $O_\theta(z) = \{y = (y_1, \dots, y_n) : |y_i - x_i| < \theta, i = 1, \dots n\}$. The probability $P_k(z)$ for hitting test point z_k into $O_\theta(z)$ regarding a given reference point w_k is expressed as follows

$$P_k(z) = \prod_{i=1}^{n} A_{ik} \int_{x_i-\theta}^{x_i+\theta} \exp(-(w_{ik} - x_{ik})^2/\sigma_k) dx_{ik}.$$

Taking into account (2) and (3), we can make the following estimates

$$A_{ik} > \frac{1}{\sqrt{\pi\sigma_k}} = \frac{\sqrt{\ln k}}{D\sqrt{2n\pi}};$$

$$1 > \exp(-\frac{(w_{ik} - x_{ik})^2}{\sigma_k}) > \exp(-\frac{D^2 \ln k}{2nD^2}) = \exp(-\frac{\ln k}{2n}).$$

$$P_k(z) > \prod_{i=1}^{n} A_{ik} 2\theta \exp(-\frac{\ln k}{2n}) > (2\theta \frac{\sqrt{\ln k}}{D\sqrt{2\pi}})^n (\frac{1}{\sqrt{k}}) = (\frac{\theta\sqrt{2\ln k}}{D\sqrt{\pi}})^n (\frac{1}{\sqrt{k}}).$$

For any cell, there exists $O_\theta(z)$ that is a subset of this cell. Then the probability getting the test point z_k into the cell is grater then $P_k(z)$.The probability Q_N of not getting a single test point into the selected neighborhood during N iterations is estimated as follows

$$Q_N < (1 - (\frac{\theta\sqrt{2\ln N}}{D\sqrt{\pi}})^n (\frac{1}{\sqrt{N}}))^N. \tag{5}$$

The probability Q_{NM} of not hitting a test point in N steps in at least one cell can be estimated as follows $Q_{NM} < MQ_N$. Consequently, in N steps all cells will be represented by at least one test point with a probability greater than (4), which is what needed to be proven.

The resulting estimate allows us to calculate the required number of steps based on the given diversity. Thus, we can form a stopping condition, which consists in achieving a given diversity with a given probability, and convert this condition into a limit on the number of steps. This allows us to obtain a specified amount of information about the location of the global maximum (Fig. 1).

Figure 1 shows the distribution of the SoFA test points across 40 subintervals of the search space [–1,1] depending on steps k for the objective function $F(z) = (1/4)\sin(4\pi z^2)\ln(z + 3/2)(((1.2)z + 3/4)^3 + 1) + 1$. It can be seen that every subinterval contains at least one test point at the 2000-th step.

Consequence. *Under the assumptions made, the sequence of test points generated by the SoFA is everywhere dense in Π with probability one for an unlimited increase in the number of steps k.*

Proof. In Theorem 1, it was shown that the probability Q_N of not getting a single test point into the chosen neighborhood during N iterations of the method is estimated by the inequality (5). Since

$$\lim_{N\to\infty} (1 - (\frac{\theta\sqrt{2\ln N}}{D\sqrt{\pi}})^n (\frac{1}{\sqrt{N}}))^N = 0,$$

then $Q_N \to 0$ when $N \to \infty$. But this also means that the probability of at least one test point falling into $O_\theta(w)$ tends to one. Due to the arbitrariness of the choice of w and θ, at least one test point will fall into any arbitrarily small neighborhood of any point in Π, which is what needed to be proven.

It can be noted that achieving a given diversity will be achieved most quickly by the Monte Carlo random search method with a uniform distribution density

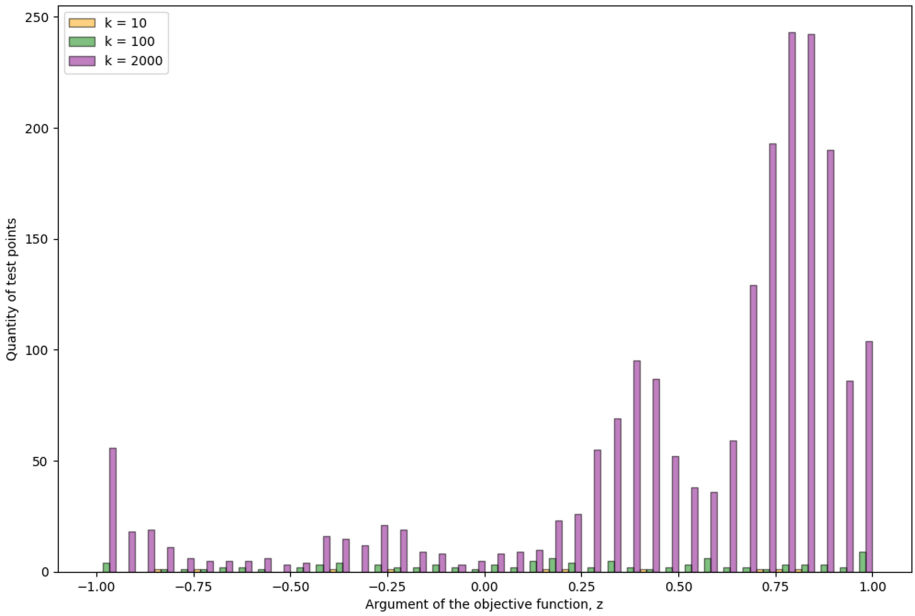

Fig. 1. Distribution of the quantity of SoFA test points across 40 subintervals of the search space [–1,1] depending on steps k for the objective function $F(z) = (1/4)\sin(4\pi z^2)\ln(z+3/2)(((1.2)z+3/4)^3+1)+1$.

of test points throughout the search space. The SoFA method in the first steps of implementation, with large values of variance σ_k and values of the exponent $\psi(k)$ close to zero, manifests itself similarly to the uniform random search method. This property can be seen in other bioinspired methods. As a rule, the formation of the initial population is carried out by the method of uniform random scattering of points throughout the search space. This behavior is justified by the need to quickly achieve the required diversity and the objectives of the space review process.

3.3 Stopping Condition

Bioinspired methods, unlike the Monte Carlo method, increase the efficiency of optimization using the second process - exploitation. This leads to a gradual concentration of test points near the best point. Let us show that test points in the SoFA method are concentrated over time in an arbitrarily small neighborhood of the global maximum of the objective function.

Theorem 2. *Let condition (3) be satisfied and the objective function have an unique global maximum point in Π. Then the probability of choosing a point for the next test from an arbitrarily small neighborhood of the global maximum point tends to one with an unlimited increase in the number of steps with probability one.*

Proof. Let $z^* = (x_1^*, \ldots, x_n^*)$ be the single global maximum point of the function $F(z)$ in Π, δ be an arbitrarily small positive number. Consider the cubic neighborhood $O_\delta(z^*)$ of the point z^*. Let us show that over time the probability of choosing a reference point w_k from $O_\delta(z^*)$ will tend to one. Let V be the complement of $O_\delta(z^*)$, $F' = sup_V F(z)$. Obviously $F' < F(z^*) = F^*$. Let us denote $\epsilon = F^* - F'$. Due to the continuity of the function $F(z)$, there is a cubic θ-neighborhood $O_\theta(z^*) \subset O_{\delta/2}(z^*)$ of the point z^* such that for any z in this neighborhood the inequality $F(z) > F^* - \epsilon/2$ holds. Since the sequence z_k is everywhere dense in Π with probability one, there exists a number m such that $z_m \in O_\theta(z^*)$. Let's consider an arbitrary k-th step under the condition $k > m$. A point can be selected as a reference point at the k-th step if it is included in the population at this step. Let point z_j be included in the population and not fall into $O_\theta(z^*)$. From condition (1) it follows that point z_m is also included in the population. The following estimate is valid for the probability $p(j,k)$ of choosing point z_j as a reference point w_k: $p(j,k) < (F')^{\psi(k)}/(F^* - \epsilon/2)^{\psi(k)}$. Then the probability of choosing an arbitrary point from V as a reference can be estimated from above by the following expression $k(F')^{\psi(k)}/(F^* - \epsilon/2)^{\psi(k)}$. Its value tends to zero as the number of steps increases indefinitely. Consequently, the probability of choosing a reference point w_k from the neighborhood $O_\theta(z^*)$ tends to one as $k \to \infty$. If $w_k \in O_\theta(z^*)$, then the probability $P_k(z^*)$ of choosing the next test point z_k from the neighborhood $O_\delta(z^*)$ can be estimated from below as follows

$$P_k(z^*) > \prod_{i=1}^{n} \int_{x_i^*-\delta}^{x_i^*+\delta} \frac{\exp(-\frac{(w_{ik}-x_{ik})^2}{\sigma_k})}{\sqrt{\pi\sigma_k}} dx_{ik} = \prod_{i=1}^{n} \int_{q_k}^{s_k} \frac{\exp(-y^2/2)}{\sqrt{2\pi}} dy, \qquad (6)$$

where $q_k = (x_i^* - w_{ik} - \delta)\sqrt{\ln k}/D\sqrt{n}$, $s_k = (x_i^* - w_{ik} + \delta)\sqrt{\ln k}/D\sqrt{n})$.

Since $x_i^* - \delta < x_i^* - \theta < w_{ik} < x_i^* + \theta < x_i^* + \delta$, then $q_k \to -\infty, s_k \to +\infty$ if $k \to +\infty$. This means that the right side of estimate (6) increases to one, therefore, the probability of choosing the point of the next test from an arbitrarily small neighborhood $O_\delta(z^*)$ of point z^* tends to one as the number of steps increases, which is what needed to be proved.

Figure 2 shows a graph of changes in the probability distribution of choosing the next test point as the number of steps increases. The graph shows that the probability distribution becomes concentrated in an arbitrarily small vicinity of the global maximum point over time. Figure 3 shows that the test point concentrate at the subinterval containing the best point over time. Here $\psi(k) = [k/10] + 1$.

The selected subset of the possible location of the global maximum point is most appropriately considered as fuzzy. The probability of choosing the next test point can be taken as an indicator of membership in this subset. As the number of steps increases, the degree of subset selection increases. If we consider some clear subset in the search space, then the degree of its selection will be equal to the probability of the point of the next test falling into it. The degree of its selection can also be characterized by the proportion of test points that fell into this subset of their total number.

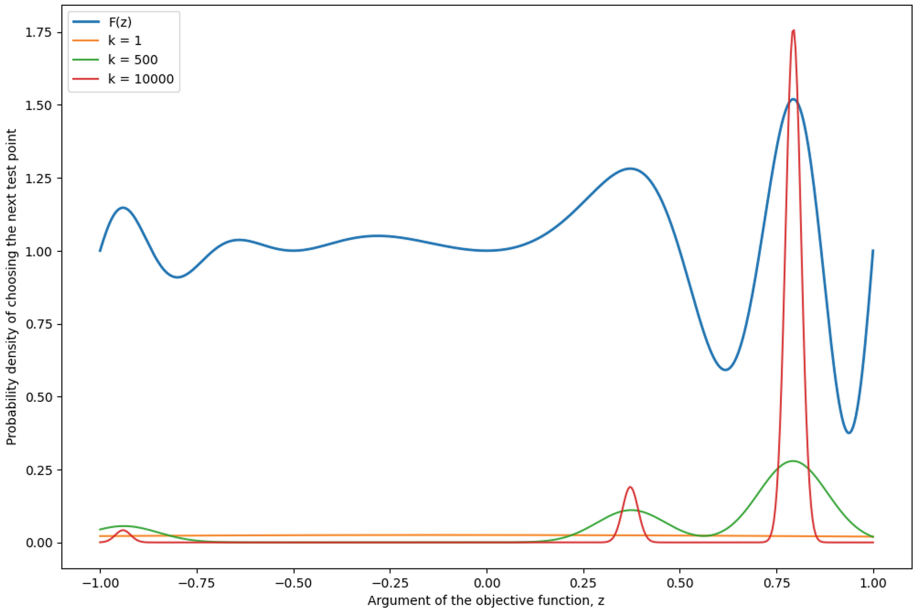

Fig. 2. Change in the probability distribution of choosing the next test point when finding the maximum of the function $F(z) = (1/4)\sin(4\pi z^2)\ln(z + 3/2)(((1.2)z + 3/4)^3 + 1) + 1$ specified on the segment [–1, 1] depending on the method step number k

Now we can give a new condition for stopping the method. For predetermined small positive numbers ϵ and δ, it is necessary to continue the search procedure until the membership index for some δ-neighborhood does not exceed the level $1-\epsilon$. Such a condition will guarantee that the constructed approximation will differ from the real maximum point by more than δ with a reliability no lower than $(1-\epsilon)$. This condition actually means obtaining a given amount of information about the location of the global maximum point.

Thus, the criteria for stopping the method are the following conditions. Firstly, it is achieving the desired diversity. Achieving the desired diversity with a plurality of test points indicates a sufficient overview of the space and the acquisition of a sufficient supply of information about the behavior of the target function in a given space. Secondly, this is the achievement of given points in some local area of a specified size. This indicates the identification of such an area as the location of the global extremum point.

Note that, unlike deterministic algorithms, obtaining a given probability is guaranteed in the stochastic case only with a certain probability. There is always some possibility of error. However, as follows from the proven theorems, for the SoFA method the probability of error decreases to zero with an unlimited increase in the number of steps.

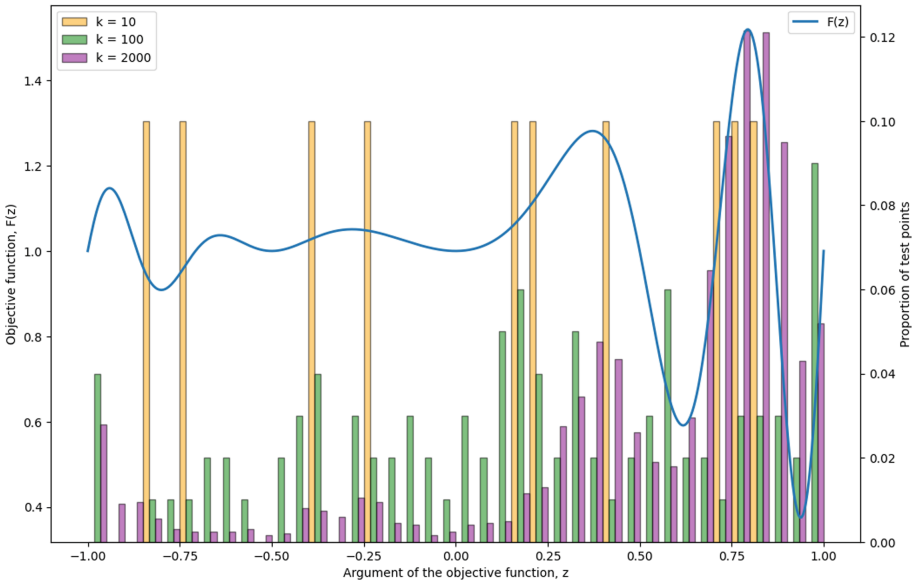

Fig. 3. Distribution of the shares of test points over 40 subintervals when finding the maximum of the function $F(z) = (1/4)\sin(4\pi z^2)\ln(z + 3/2)(((1.2)z + 3/4)^3 + 1) + 1$ on the segment [–1, 1] depending on the SoFA step number k

The significance of the new definition of information is not limited to the area of optimization. Most modern information technologies are built on the basis of the use of optimization procedures, for example, training neural networks, solving problems of classification, ranking, etc. [35,36]. Accordingly, a new understanding of information can be useful here too.

The information analysis of optimization processes was introduced into the educational process at Lobachevsky State University of Nizhny Novgorod for training bachelors of the studying area "Fundamental Informatics and Information Technologies" when studying the disciplines "Bio-inspired optimization methods" and "Mathematical modeling of selection processes." These disciplines are optionally studied by students in the third year of study. Here a theoretical understanding of the essence of information is applied to the information analysis of global optimization methods and helps improve these methods. The acquired knowledge is consolidated in practice.

As the experience of teaching students shows, the most effective method of practical development of professional skills is the project approach. Taking into account this fact, project-based teaching methods are used here. The project methodology used is described in detail in [6]. Students carry out project work on the topics "Searching for optimal settings of the SoFA method", "Modifying the SoFA method to increase optimization efficiency", 'Parallel implementation of the SoFA method", "Using the SoFA method when training a neural network" and others. To carry out projects, the algorithmic language Python is used with

numpy plugin combined with an extensive library of high performance mathematical functions. Parallel implementation of the SoFA method ensures simultaneous independent calculations of the objective function values. Carrying out project work inevitably motivates students to develop an in-depth understanding of the theoretical concept of information and its implications for practice. Thus, when optimizing the method settings and selecting the best dependence $\psi(k)$, the student has to independently evaluate the diversity of the set of tests generated by the method with different settings, and evaluate the information about the global optimum obtained for a given step. The illustrative graphs included in this article were prepared by student S.A. Nalchajyan during his project work. The experience of introducing a new approach to information into the educational process shows that it gives students a clearer and deeper understanding of many fundamental methods of information technology and allows them to more consciously improve them.

4 Conclusion

As a result of the study, a new understanding of the concept of information in the area of information technology was proposed. It is shown how this approach can be used in the educational process of training IT specialists. The correct definition of the concept of information has methodological significance in the formation of digital culture. It allows us to form clear ideas about the essence of the basic concept, thanks to which the training system becomes logical and consistent. This definition is not strictly linked to probability theory and allows us to overcome the shortcomings of previous approaches. This definition can be used to improve numerical optimization methods, as well as other information technology methods.

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